

For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm in python to output a description of the set of all hypotheses consistent with the training examples

code

```
import pandas as pd

import numpy as np

# -----

# Create Training Data and Save as CSV

# -----

data = {

    "Sky": ["Sunny", "Sunny", "Rainy", "Sunny"],

    "AirTemp": ["Warm", "Warm", "Cold", "Warm"],

    "Humidity": ["Normal", "High", "High", "High"],

    "Wind": ["Strong", "Strong", "Strong", "Strong"],

    "Water": ["Warm", "Warm", "Warm", "Cool"],

    "Forecast": ["Same", "Same", "Change", "Change"],

    "EnjoySport": ["Yes", "Yes", "No", "Yes"]

}

df = pd.DataFrame(data)

df.to_csv("trainingdata.csv", index=False)

# -----

# Read CSV File

# -----

dataset = pd.read_csv("trainingdata.csv")
```

```

print("Training Dataset:\n")
print(dataset)

# -----
# Separate Features and Target
# -----
concepts = np.array(dataset.iloc[:, :-1])
target = np.array(dataset.iloc[:, -1])

# -----
# Candidate Elimination Algorithm
# -----
specific_h = concepts[0].copy()

general_h = ["?" for i in range(len(specific_h))]
    for i in range(len(specific_h))

print("\nInitial Specific Hypothesis:")
print(specific_h)

print("\nInitial General Hypothesis:")
for row in general_h:
    print(row)

# Training
for i, h in enumerate(concepts):

    if target[i] == "Yes":

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for x in range(len(specific_h)):
    if h[x] != specific_h[x]:
        specific_h[x] = "?"
        general_h[x][x] = "?"

elif target[i] == "No":
    for x in range(len(specific_h)):
        if h[x] != specific_h[x]:
            general_h[x][x] = specific_h[x]
        else:
            general_h[x][x] = "?"

print("\nStep", i + 1)
print("Specific Hypothesis:", specific_h)
print("General Hypothesis:")
for row in general_h:
    print(row)

# Remove duplicate general hypotheses
general_h = [g for g in general_h if g != ['?', '?', '?', '?', '?', '?']]

print("\n=====")
print("Final Specific Hypothesis (S):")
print(specific_h)

print("\nFinal General Hypothesis (G):")
for row in general_h:
    print(row)

```

Training Dataset:

	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
0	Sunny	Warm	Normal	Strong	Warm	Same	Yes
1	Sunny	Warm	High	Strong	Warm	Same	Yes
2	Rainy	Cold	High	Strong	Warm	Change	No
3	Sunny	Warm	High	Strong	Cool	Change	Yes

Initial Specific Hypothesis:

['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']

Initial General Hypothesis:

['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']

Step 1

Specific Hypothesis: ['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']

General Hypothesis:

['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']

Step 2

Specific Hypothesis: ['Sunny' 'Warm' '?' 'Strong' 'Warm' 'Same']

General Hypothesis:

['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']

Step 3

Specific Hypothesis: ['Sunny' 'Warm' '?' 'Strong' 'Warm' 'Same']

General Hypothesis:

['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', 'Same']

Step 4

Specific Hypothesis: ['Sunny' 'Warm' '?' 'Strong' '?' '?']

General Hypothesis:

['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']

=====

Final Specific Hypothesis (S):

['Sunny' 'Warm' '?' 'Strong' '?' '?']

Final General Hypothesis (G):

['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']